A REALLY GOOD TITLE

Paxton Fitzpatrick

April 16, 2019

1 Introduction

2 Methods

2.1 Experimental design and data collection

The experiment was designed and implemented via the PsiTurk platform (Gureckis et al., 2015), and coded in a combination of JavaScript (specifically, the JsPsych library; de Leeuw, 2015) and Python. The study took place over two one-hour sessions, separated by six to eight days (mean 6.7 days; median 6.9 days) to approximate the common time span between releases of subsequent television episodes. Participants (N = 70) were divided into two equal groups. In the first session, all participants viewed the 25-minute pilot episode of the FX series, *Atlanta*. Following the episode, participants were then given up to an hour to recall its events, in order, to the best of their abilities. Mirroring Chen et al. (2017) , they were asked to try to speak for a minimum of ten minutes and told that they were free to go back and describe earlier events, should they realize they missed any. Finally, they were given up to five minutes to predict what they thought would happen in the next episode. In the second session, all participants were again given up to one hour to recall to recall the episode viewed the prior week. Group one then viewed the second episode of *Atlanta*, while group two viewed the pilot of *Arrested Development*. Both groups finished by recalling the newly watched episode in the same manner. Eleven subjects' data were discarded prior to processing and analysis: five did not follow the provided instructions, two did not complete the task, two did not return for the second session, one dropped out due to illness, and one was unable to complete the experiment due to technical issues. The data analyzed comprise the remaining 59 participants (30 in group one; 29 in group two).

2.2 Stimulus annotation and audio transcription

The ability of Heusser et al. (2018)'s approach to model the explicit content of a naturalistic experience relies on the availability of detailed, hand-generated annotations that characterize the experience's evolving content. To create the annotation corpus for each television episode, I used ANVIL (Kipp, 2001), along with a custom XML specification file to record the following features for every camera shot:

- A brief description of physical occurrences and characters' actions (*"Narrative details: external events"*)
- A brief description of characters' emotional states and various subtextual implications (*"Narrative details: internal states"*)
- A list of characters appearing on-screen
- A transcription of all dialogue
- The name of the character speaking
- The shot's setting

- Whether the shot took place indoors or outdoors
- The presence or absence of music

These features were chosen both to be consistent with and to improve upon prior work with a similar set of stimulus annotation generated by Chen et al. (2017). Analyses carried out on Chen et al.'s feature set in conjunction with Heusser et al.'s modeling approach revealed that certain annotated features (e.g., what text appears on screen) contributed little-to-no structure to the stimulus trajectory. Further, some features (e.g., name of the character in focus and of the character speaking) were found to be highly redundant with each other, while others (e.g., the camera angle) were rarely preserved during recall and thus contributed unnecessary noise to the model. I chose to separate descriptions of physical actions from internal emotions and subtextual implications both to acknowledge that not all content is presented equally explicitly, and to enable future investigations by myself or others into potential differences in prioritization of those features in processing and memory.

I transcribed all participant audio using the Google Cloud Speech-to-Text API (**?**). A previous proof-of-concept study (**?**) suggested that the Google's automatic transcription API in particular may be used as a

2.3 Dynamic content modeling, event segmentation, and video-recall mapping

3 Analyses to be employed and expected results

3.1 Comparing immediate and delayed recall

3.2 Measuring the effect of contextual familiarity on future memory success

3.3 Relating prediction ability to memory success

4 Implications

References

Chen, J., Leong, Y. C., Norman, K. A., and Hasson, U. (2017). Shared experience, shared memory: a common structure for brain activity during naturalistic recall shared experience, shared memory: a common structure for brain activity during naturalistic recall. *Nature Neuroscience*, 20(1):115–125.

de Leeuw, J. R. (2015). jsPsych: A JavaScript library for creating behavioral experiments in a web browser. *Behavior Research Methods*, 47(1):1–12.

Gureckis, T. M., Martin, J., McDonnell, J., Rich, A. S., Markant, D., Coenen, A., Halpern, D., Hamrick, J. B., and Chan, P. (2015). psiTurk: An open-source framework for conducting replicable behavioral experiments online. *Behavior Research Methods*, 48(3):829–842.

Heusser, A. C., Fitzpatrick, P. C., and Manning, J. R. (2018). How is experience transformed into memory? *bioRxiv*, 409987.

Kipp, M. (2001). Anvil-a generic annotation tool for multimodal dialogue. In *Seventh European Conference on Speech Communication and Technology*.